EXPERIMENT – 7

AIM :

To write a Python program to implement Breadth-First Search (BFS) to traverse a graph level by level.

ALGORITHM:

1. Start.
2. Initialize a queue and a visited list.
3. Insert the starting node into the queue and mark it as visited.
4. Remove the front node from the queue and display it.
5. Add all unvisited adjacent nodes to the queue and mark them as visited.
6. Repeat steps 4 and 5 until the queue becomes empty.
7. Stop.

PROGRAM :

untitled (3.14.8)

File Edit Format Run Options Window Help

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [], 'E': [], 'F': []
}

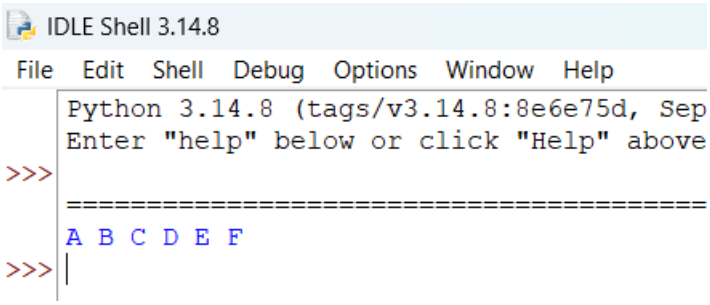
def bfs(start):
    visited = {start}
    q = deque([start])

    while q:
        node = q.popleft()
        print(node, end=" ")

        for n in graph[node]:
            if n not in visited:
                visited.add(n)
                q.append(n)

bfs('A')
```

OUTPUT:



RESULT:

Thus, the Python program to implement BFS was executed successfully, and the graph nodes were traversed level by level.